

Continuous Core Body Temperature Monitoring for Heatstroke Alert via a Wearable In-Ear Thermometer

Published as part of ACS Sensors special issue "Wearable Sensors".

Wei Zhang,^{||} Lei Li,^{||} Yimeng Wang,^{||} Xue Dong,^{*} Cenyu Liu, Liyang Sun, Qinglin Guan, Feng Zhang, and Shuogui Xu^{*}



Cite This: ACS Sens. 2025, 10, 1440–1449



Read Online

ACCESS |



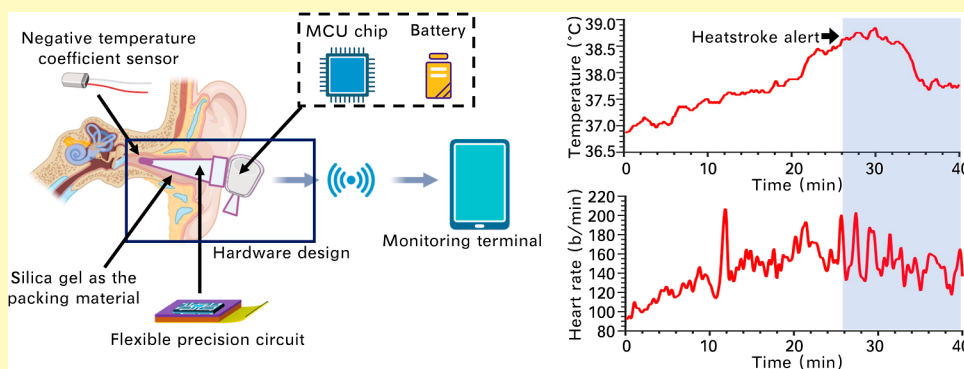
Metrics & More



Article Recommendations



Supporting Information



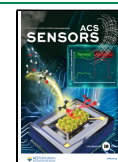
ABSTRACT: Heatstroke, a global concern exacerbated by climate change, poses significant health risks, potentially leading to multiorgan damage and fatalities. Core body temperature (CBT) is a critical and precise indicator of heatstroke, and its continuous monitoring could serve as a pivotal tool for early detection. Traditional CBT measurements, often invasive (e.g., surgical intubation, rectal or oral placement), and disrupt daily activities, whereas existing wearable devices predominantly measure skin temperatures which is susceptible to ambient environment, thus unreliable for heatstroke alert. To overcome these limitations, this study introduces an innovative in-ear wearable device to measure CBT via the cochlea, which allows for accurate CBT monitoring and timely heatstroke alerts during activities in high-temperature and high-humidity environments. The device comprises a negative temperature coefficient (NTC) thermometer integrated into a flexible precision circuit (FPC), a compact Bluetooth module, an 8 mA h micro battery, and a biocompatible, low-stimulus silica gel casing. With dimensions of 27 mm × 18 mm and a maximum in-ear diameter of 5 mm, weighing just 1.3 g, the device offers high portability and comfort, with a continuous operational lifespan of at least 24 h postcharging. A complementary software system facilitates continuous CBT monitoring, heatstroke alerts, and device management. Preliminary human trials demonstrate the device's accuracy in CBT measurement, rivaling that of rectal thermometry, and superior to the performance of surface body temperature measurement at different body parts. Long-term experiments affirm the device's efficacy in detecting rapid CBT escalations, enabling timely preventive measures against heatstroke.

KEYWORDS: heatstroke, core body temperature, wearable device, negative temperature coefficient thermometer, in-ear thermometer

Heatstroke, a critical medical emergency characterized by a rapid elevation of core body temperature (CBT) beyond homeostatic thresholds (typically >40.5 °C), precipitates a cascade of detrimental physiological effects, including reduced cardiac output, compromised thermoregulatory mechanisms, central nervous system impairment, organ failure, and ultimately, mortality.^{1–3} This condition poses significant risks across diverse populations, with classical heatstroke (CHS) arising from environmental exposure disproportionately affecting those with limited physiological adaptability, such as the young, elderly, and individuals with pre-existing health issues. Conversely, exertional heatstroke (EHS), linked to strenuous physical activity under high-temperature environ-

ments, predominantly impacts athletes and outdoor laborers.^{4–6} In the context of global warming, a substantial 31% of 136 global regions have witnessed unprecedented extreme temperatures over the past six decades, exacerbating the challenge of heat-related health threats.^{7,8} The "Lancet

Received: December 6, 2024
Revised: December 24, 2024
Accepted: January 13, 2025
Published: January 21, 2025



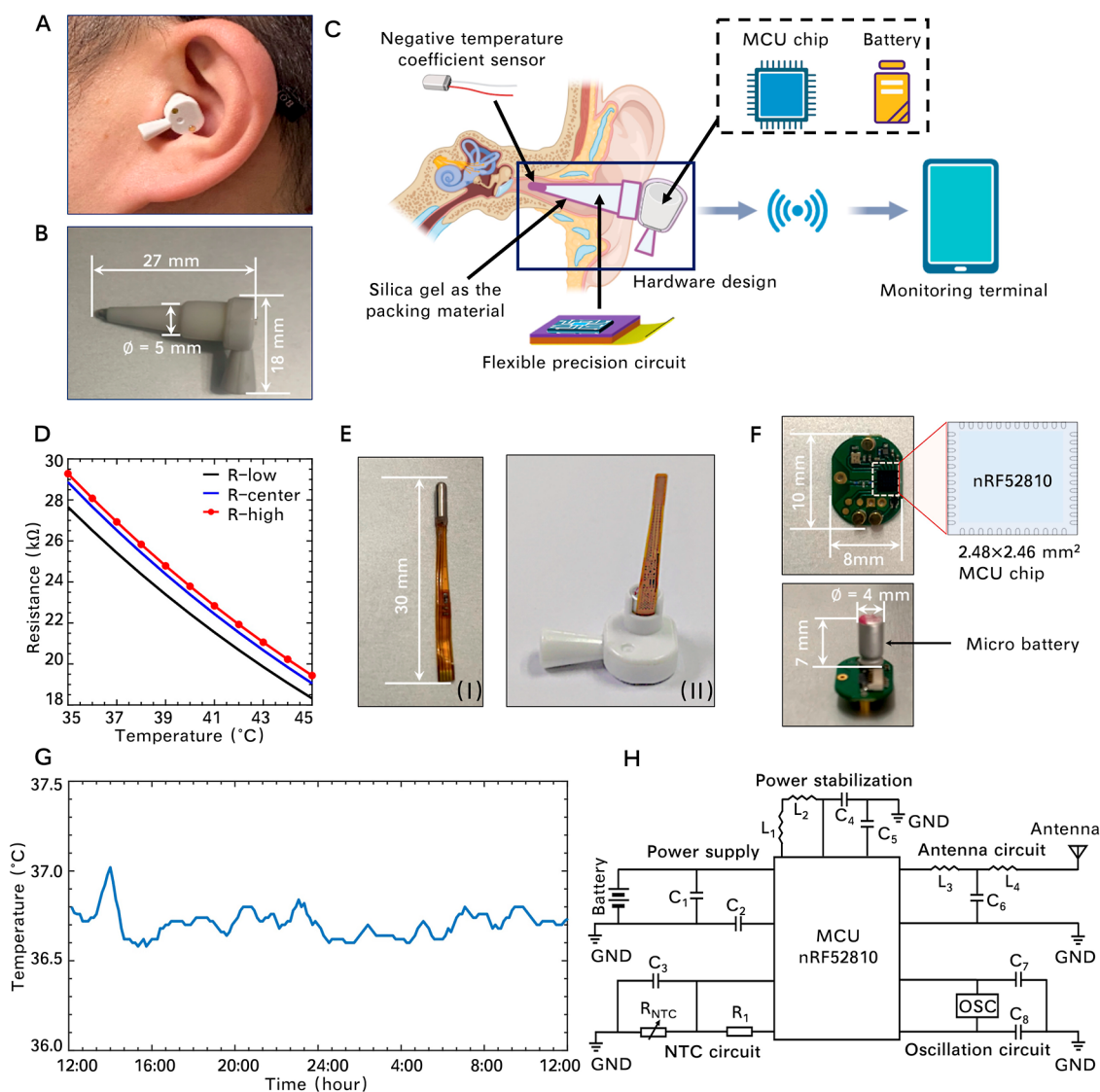


Figure 1. (A) Image of a participant wearing the proposed in-ear thermometer; (B) the overall appearance and size of the device; (C) the schematic diagram of the whole design; (D) resistance as a function to the temperature of the NTC sensor; (E) the flexible precision circuit (I) with an NTC sensor embedded and (II) connected with the rest of the circuit; (F) the miniaturized board with micro Bluetooth MCU chip and micro battery embedded; (G) 24 h monitoring record of the in-ear thermometer; and (H) the hardware architecture.

Countdown on Health and Climate Change” reports a marked escalation in heatwave exposure and global heat-related mortality in recent years.⁹ Meanwhile, an analysis spanning 1990 to 2019 indicates that heatwave-induced excess mortality is a consistent threat across all continents.¹⁰ Alarmingly, the relentless increase in global temperatures could lead to a 3.7-fold rise in heat-related deaths by 2050.¹¹ Consequently, taking effective measures to prevent heatstroke, one of the most serious heat-related illnesses, is of great urgency to the clinical community.

The timely detection and identification of heatstroke are paramount within the spectrum of intervention strategies.^{12–14} Core body temperature (CBT) is inextricably linked to the pathology of heatstroke and is recognized as the most direct and reliable diagnostic metric.^{15,16} Traditional CBT measurement methods, involving invasive procedures such as intrapulmonary cannulation or rectal and oral implants, are not only intrusive but also cumbersome, posing significant challenges to the facile and continuous monitoring of CBT.¹⁷ In recent

years, wearable devices have emerged as promising tools, offering enhanced portability and endurance.^{18,19} These devices have been engineered to capture a range of physiological signals, including heart rate,¹⁵ skin temperature,^{20–22} and sweating,^{23,24} as alternatives for CBT to facilitate early heatstroke detection. Despite their innovative approach, these wearable technologies, which rely on indirect and skin-based measurements, still exhibit limitations in accuracy and susceptibility to environmental interference. Especially during activities in high-temperature and high-humidity environments, which are often associated with heatstroke, the presence of wet skin, excessive perspiration and heat transfer can interfere with signal transmission through surface body sensors. This can lead to unreliable monitoring and failure to trigger heatstroke alerts.

The cochlea, recognized as a viable locus for CBT measurement due to its protective ear canal structure, presents a promising avenue for long-term monitoring with the aid of wearable sensor technology. This anatomical feature, charac-

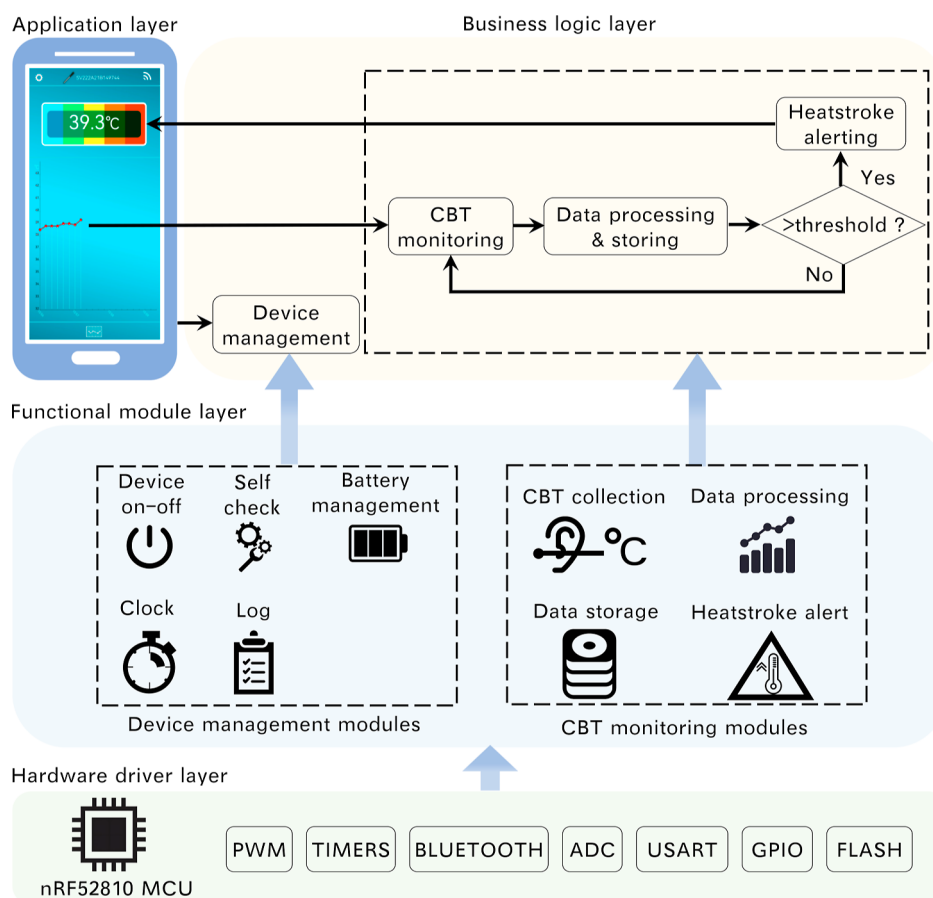


Figure 2. Schematic of the software system.

terized by its essentially open configuration and low sensitivity to minimally invasive sensors, has inspired a handful of pioneering investigations.^{25–27} However, the devices developed in these studies, such as those measuring 58 mm × 58 mm × 21 mm²⁵ and 45 mm × 38 mm × 18 mm,²⁶ respectively, are not sufficiently compact or comfortable for prolonged daily use and continuous CBT monitoring. Thus, there remains a critical need for advancements in wearable technology that can seamlessly integrate with daily life while providing accurate and reliable CBT data.

This study introduces an innovative in-ear thermometer, meticulously miniaturized in both dimensions and mass, designed to facilitate continuous CBT monitoring and heatstroke detection with unprecedented flexibility and wearer comfort. The device, the most compact of its class to date, integrates a diminutive thermistor, a flexible embedded circuit, a miniature Bluetooth module, and a compact battery, culminating in an overall size of 27 mm × 18 mm, with a maximum in-ear diameter of 5 mm. Extensive human trials have validated the device's performance, demonstrating a high degree of accuracy in CBT measurement that rivals that obtained via rectal thermometry. Furthermore, the device exhibits robust capabilities in the continuous surveillance of CBT and the timely issuance of heatstroke alerts, underscoring its potential as a transformative tool in the realm of wearable health technology.

MATERIALS AND METHODS

Overall Design. Figure 1 presents the hardware architecture of the proposed in-ear CBT monitoring device. As depicted in Figure

1A,B, the device adopts an earphone-shaped design, measuring 27 mm in length and 18 mm in height, with a maximum in-ear diameter of 5 mm. Figure 1C delineates the schematic configuration of the entire device, comprising a tiny negative temperature coefficient (NTC) thermistor integrated within an elongated flexible circuit to accurately sense CBT at the cochlea, a compact Bluetooth module for the wireless transmission of CBT data to a monitoring interface, and a miniature battery designed for an 24 h operational duration. The device is encased in lightweight, 3D-printed material and biocompatible silica gel, which minimizes discomfort and irritation within the ear canal.

Designs of the Hardware. To ensure portability, the proposed device incorporates four specific hardware designs. First of all, the NTC thermistor was utilized as the temperature sensor owing to its superiority in rapid response (within 3 s) and high precision ($\pm 1\%$). This thermistor measures voltage variations via an RC (resistor capacitance) impedance integration circuit, which includes a high-precision reference resistor, and subsequently converts these readings to temperature using the resistance–temperature characteristic curve illustrated in Figure 1D. To accommodate the diverse geometries of human ear canals while ensuring the NTC circuit reaches the cochlea, an elongated FPC (flexible precision circuit) with a thickness of 0.3 mm was adopted to embed the NTC thermistor and its circuits as is shown in Figure 1E. Second, the advanced micro SoC (system on a chip) chip, i.e., the nRF52810 designed by Nordic Semiconductor (Norway) and manufactured by TSMC (Taiwan), was specifically utilized as the MCU (microprogrammed control unit) to manage the device, process the signals, as well as transmit CBT to monitoring terminals. This chip, equipped with a 32 MHz ARM Cortex-M4 processor, supports ultralow energy consumption during Bluetooth communication and boasts a compact footprint of 2.48 mm × 2.46 mm. Additionally, a micro battery with dimensions of 7 mm in height and 4 mm in diameter was selected. As depicted in Figure 1F, the

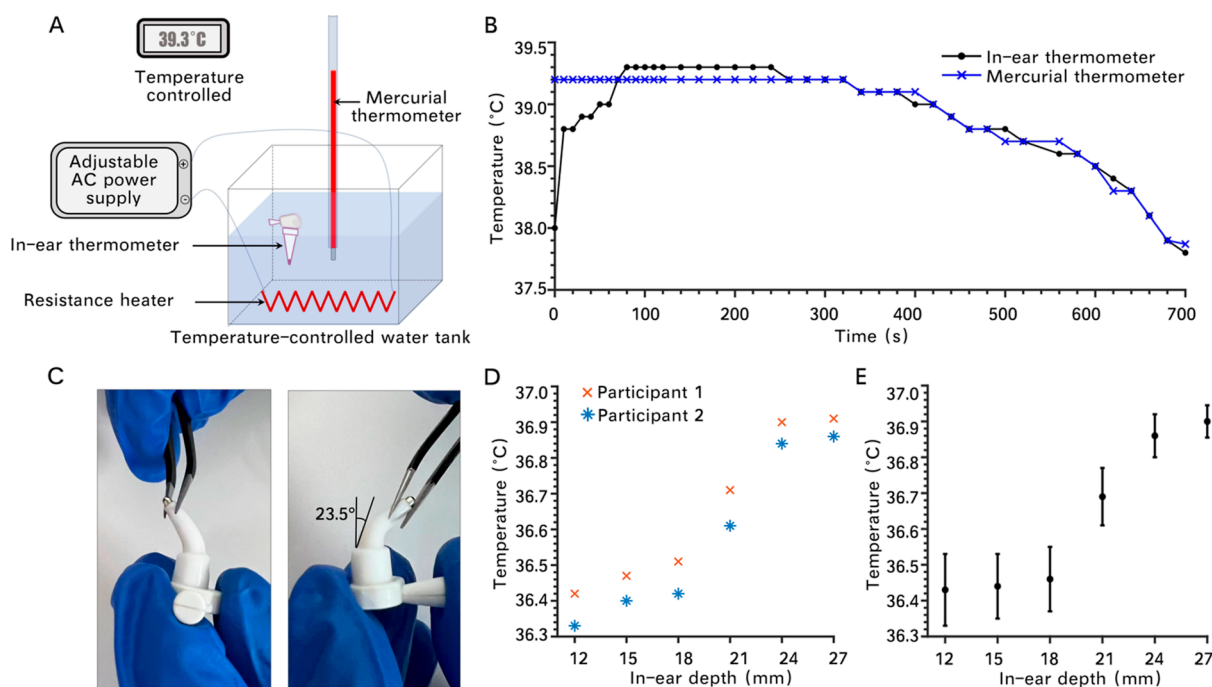


Figure 3. Calibration and repeatability test of the in-ear thermometer. (A) Experimental setups for the self-calibration of the in-ear thermometer; (B) temperature variation during calibration of both the in-ear thermometer and the mercurial thermometer; (C) flexibility test of the in-ear component made of flexible precision circuit (FPC) encased in soft silica gel; (D) temperature and its variation at different in-ear depths from two distinct participants; and (E) averaged temperature and its variation at different in-ear depths, data were collected from six repeated experiments with five volunteers involved.

MCU and battery are ingeniously integrated onto a single mini-board measuring just 8 mm × 10 mm, thereby minimizing the overall device size and significantly reducing its weight. The adopted battery can provide a continuous power supply for over 24 h based on the device's power consumption (see [Supporting Information](#)), which is also practically demonstrated in a monitoring record shown in [Figure 1G](#). [Figure 1H](#) presents a detailed outline of the hardware architecture. This architecture is designed to bring together multiple crucial circuits into a single, unified system. These circuits include the battery circuit, which is responsible for power supply management; the power stabilization circuit, which plays a vital role in ensuring a stable power output, thereby preventing fluctuations that could potentially disrupt the proper functioning of the device; the antenna circuit, which is essential for wireless communication; the oscillation circuit for generating periodic signals required for various internal operations within the device; and the NTC circuits, which are used for temperature sensing. All of these circuits are centrally managed by the nRF52810 MCU. It coordinates and regulates the operations of all the integrated circuits, ensuring that they work in harmony to enable the device to function as designed in data acquisition, processing, and communication. Furthermore, the device is encased in lightweight, soft, and skin-friendly 3D-printed silicone, which reduces irritation to the ear canal and contributes to the device's total weight of 1.3 g. This design not only enhances wearer comfort but also ensures the device's suitability for prolonged use in diverse environmental conditions.

Designs of the Software System. A comprehensive software system was designed in this study for long-term CBT monitoring, heatstroke alerting, data processing and storage, as well as device management, as depicted in [Figure 2](#). The application layer furnishes an intuitive and cohesive interface, facilitating user interaction with the device via their terminals (see [Supporting Information](#) materials). The logic layer defines the pipeline of the entire system, with a focus on device management that oversees the device's functionality and operational integrity. Once the device is in operation and the CBT monitoring function is activated, the system persistently acquires and stores CBT data, triggering heatstroke alerts when CBT breaches established thresholds.

To operationalize these principles, the function module layer is bifurcated into device management and CBT monitoring modules. The device management modules encompass functionalities such as device activation/deactivation, Bluetooth communication, self-diagnostic routines, battery management, timekeeping, and error logging. In contrast, the CBT monitoring modules are tasked with the collection, storage, processing, and alerting of CBT data. Additionally, the hardware driver layer ensures the seamless execution of MCU-related functions, including PWM (power management), TIMERS, Bluetooth, ADC (analog to digital converter), USART (Universal Synchronous/Asynchronous Receiver/Transmitter), GPIO (General-purpose input/output), and FLASH, thereby supporting the device's robust performance and integration.

RESULTS

Calibration. As shown in [Figure 3A](#), a self-calibrated accuracy test was conducted using a temperature-controlled water bath filled with purified water as the medium. A mercury thermometer was placed in the measurement zone to serve as the standard reference for temperature verification. The in-ear thermometer was then submerged in the water bath to perform the calibration experiment. Initially, the water was heated to a target temperature of 39.2 °C to calibrate the rising and stabilization process of the in-ear thermometer. Once the thermometer's readout stabilized, the heating was stopped, allowing the water to cool naturally, thereby enabling calibration of the cooling process. The water temperature was meticulously controlled and stabilized throughout the experiment. Should the temperature readout from the device deviate from the water bath temperature, an offset is applied to the thermometer's data processing unit to ensure that the reading aligns with the target temperature to within ± 0.1 °C. It is worth noting that the offset derived from this single temperature point is applicable across the entire operational range of the in-ear thermometer.

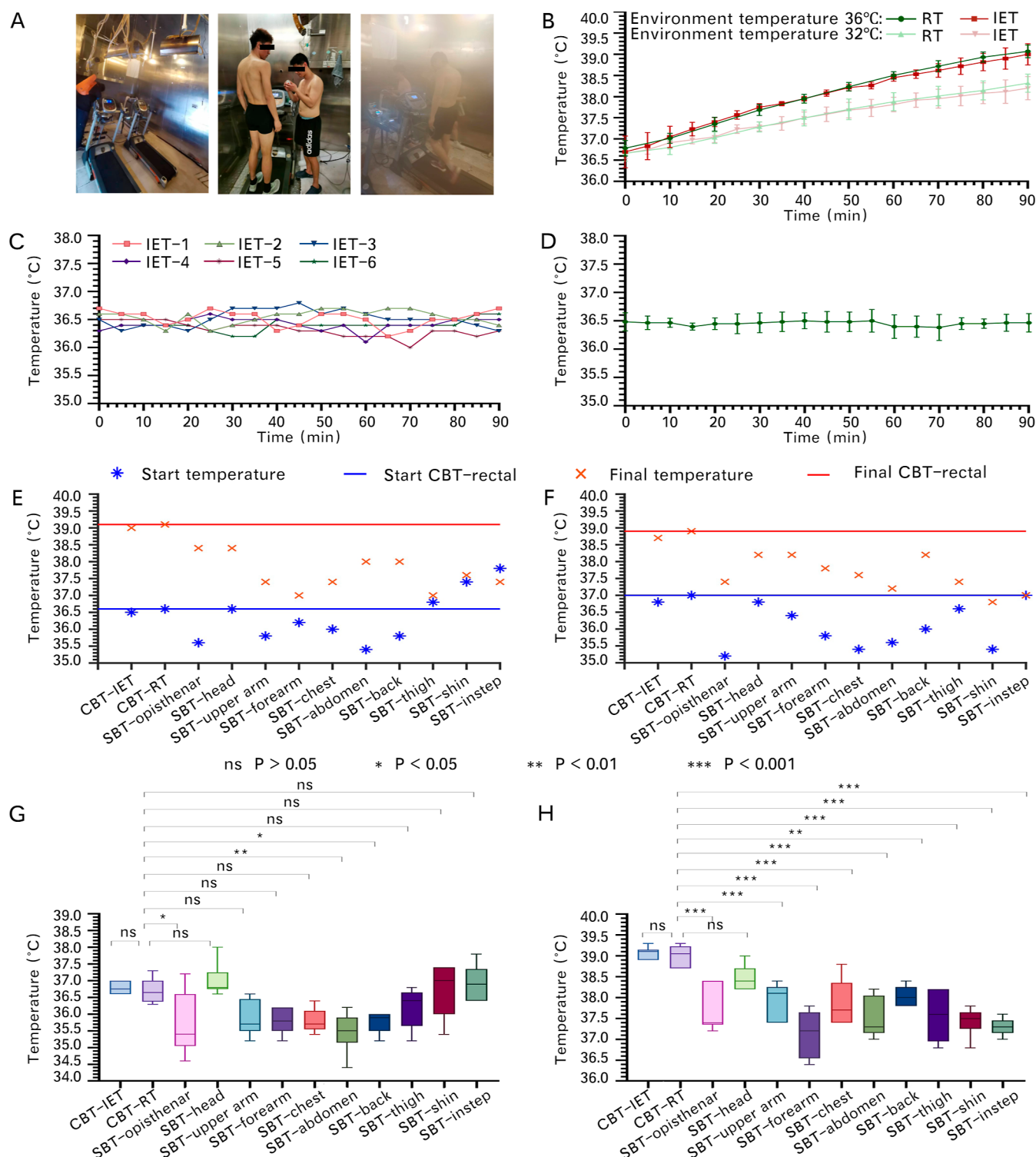


Figure 4. Performance of the in-ear thermometer. (A) Setups for the human experiment; (B) a comparison between in-ear temperature and rectal temperature averaged from 6 participants with environmental temperatures controlled at 32 and 36 °C, respectively; (C) in-ear temperatures of 6 participants at a rest state with an environmental temperature of 32 °C, and (D) depicts their arithmetic mean and standard deviations; (E,F) present the CBTs and SBTs from two randomly picked participants at the start (blue) and end (red) of the experiments in an environment temperature of 36 °C, while (G,H) show the statistical distribution of various CBTs and SBTs measured at the start and end of the experiment, with the distribution calculated from 6 participants.

The relationship between the measured temperature and the elapsed time is plotted in Figure 3B. The data demonstrate that the in-ear thermometer reaches the target temperature within 80 s, showcasing a rapid response to temperature fluctuations. Furthermore, the thermometer's readout stabilized after

reaching 39.2 °C (80–200 s), confirming its accuracy in steady-state temperature measurements. Additionally, during the cooling phase (200–700 s), the thermometer's readings closely matched those of the mercury thermometer to within

± 0.1 °C, demonstrating its accurate dynamic response during the cooling process.

Repeatability Test of the In-Ear Thermometer. Given the diverse anatomical structures of individual ear canals, experiments were conducted to assess its applicability across various ear shapes and measurement depths. As depicted in Figure 3C, the thermometer's in-ear component, comprising a flexible precision circuit encased in soft silica gel, can bend up to 23.5° , enabling it to conform to the ear canal's varied anatomical forms.

Furthermore, an experiment was carried out to gauge the thermometer's reliability at different insertion depths. Data from six independent trials revealed that at an insertion depth of 27 mm, the temperature readings exhibited the least variability. Figure 3D depicts the temperature measured at various in-ear depths from two participants, showing a similar trend but distinct readouts may be attributed to individual differences. In addition, a statistical result from five participants is shown in Figure 3E. It is important to note that the thermometer's design length of 27 mm aligns closely with the cochlea's depth, approximately 30 mm to 40 mm from the ear canal entrance,²⁸ thereby facilitating accurate cochlea temperature measurement while also maintaining wearing safety. It is also worth noting that individual variations in ear canal structure, wearing habits, and sensitivity to in-ear devices, may result in slightly shallower measurement points. For example, at a depth of 24 mm, the thermometer shows a slightly lower temperature with a similar bias of ± 0.06 °C to the 27 mm depth. However, at insertion depths starting from 21 mm, the temperature readings begin to show greater biases as the NTC sensor no longer accurately contacts the cochlea area. This bias increases at even shallower depths (e.g., 18, 15, and 12 mm), where the thermometer primarily measures middle ear temperatures and wearing stability cannot be well guaranteed.

Evaluation of CBT Measurement. A human body experiment was performed to evaluate the accuracy of CBT measurement using the proposed device. The experimental setups, detailed in Figure 4A, maintained two environmental temperature conditions, i.e., 32 and 36 °C, with a relative humidity of 60%. The study involved six participants engaging in 90 min of treadmill exercise, during which the in-ear temperature (IET) was continuously measured and recorded every 5 min, with rectal temperature (RT) taken every 10 min and served as the gold standard for CBT. Figure 4B presents the averaged IET readings in comparison to RT for the six participants. While distinct temperature trends emerged under the two environmental conditions, the measured IETs exhibit a close agreement to RTs with a maximum deviation of 0.4 °C, suggesting that the proposed device could serve as a viable alternative to RT for CBT assessment. Notably, Figure 4B reveals that between the 20th and 70th minutes, the measured IET closely matched RT, underscoring the device's efficacy and precision in monitoring CBT during physical activity, coupled with its superior portability and comfort. The observed discrepancies at the beginning and end of the exercise period may be attributed to individual physiological variations.

It is worth noting that conducting human experiments at around 40 °C and high relative humidity is rather dangerous. Since 40 °C is generally considered as a condition leading to heat stroke, experiments at such high temperature and humidity are difficult to be approved by the ethics committee. However, in a previous research,²⁹ we have conducted such

experiments in rats. We first selected a fixed relative humidity (60%) and established a gradient temperature (36–46 °C) to explore the thermoregulatory responses of rats. It was surprised to find that the plateau temperature of rats was 40 °C no matter how the environmental temperature was changed within a range of 36–42 °C. This indicates that the core body temperatures of the rats rapidly reached a plateau in the hot environment and maintained a relatively high CBT during the plateau phase. Once the rats left this plateau phase, they entered the final rising phase where heatstroke and death occurred. In addition, we also unexpectedly found that the increase in humidity did not cause significant effects on heat stress within a certain range (relative humidity 40–80%). However, when the relative humidity reaches $\geq 90\%$, the thermoregulatory response changed significantly. In particular, even at an environmental temperature of 40 °C, when the relative humidity exceeded a threshold of 90%, heat stress was observed at a level comparable to that of an environmental temperature of 44 °C. Therefore, the effect of relative humidity should also be noted at high-temperature environments. This also explains why many cases of severe heatstroke occur even in milder high-temperature environments.

In addition, a stability test involving six participants was conducted to evaluate the effects of prolonged wearing. The experiment was performed with six participants in a static state at an environmental temperature of 32 °C, thereby minimizing potential biases from dynamic CBT changes due to physical activity or high-temperature exposure. Figure 4C displays IET readings from the six participants over a 90 min period, showing minor fluctuations that reflect individual variations in body temperature. The arithmetic mean and standard deviations were shown in Figure 4D, which demonstrates a consistent trend in IET values. These findings confirm that prolonged use of the device does not accumulate heat within ear or disrupt the its thermal equilibrium, ensuring reliable and accurate temperature monitoring over extended periods.

Furthermore, the surface body temperature (SBT) at various positions, including the head, arm, forearm, chest, abdomen, back, thigh, shin, and instep, were measured using a hand-held infrared thermometer (Fluke 59, Fluke Testing Instruments (Shanghai) Co., Ltd.) at the commencement and conclusion of the experiment. This provided a comprehensive assessment and comparison of different temperature measurement methods. Figure 4E,F illustrate the distribution of CBTs and SBTs from two randomly selected participants, with rectal-acquired CBTs serving as the gold standard for both initial and final temperatures. Notably, the IETs exhibited the highest concordance with rectal thermometers, maintaining biases within 0.2 °C from start to finish. Conversely, SBTs showed varying degrees of deviation from rectal temperatures, with inconsistencies observed at the beginning and end of the experiments. This variability is attributed to the susceptibility of SBT to factors such as sweat evaporation and heat exchange with the environment, which can lead to significant divergences from core body temperature. It is important to highlight that in Figure 4E, SBTs measured at the thigh, shin, and instep were anomalously elevated compared to CBTs, underscoring the unreliability of SBT as a consistent indicator of core body temperature fluctuations. Figure 4G,H depict the statistical outcomes of IET and RT at the start and end of the experiment, alongside SBT measurements at the aforementioned anatomical sites for comparative analysis. The data reveal that IET does not significantly differ from RT at either

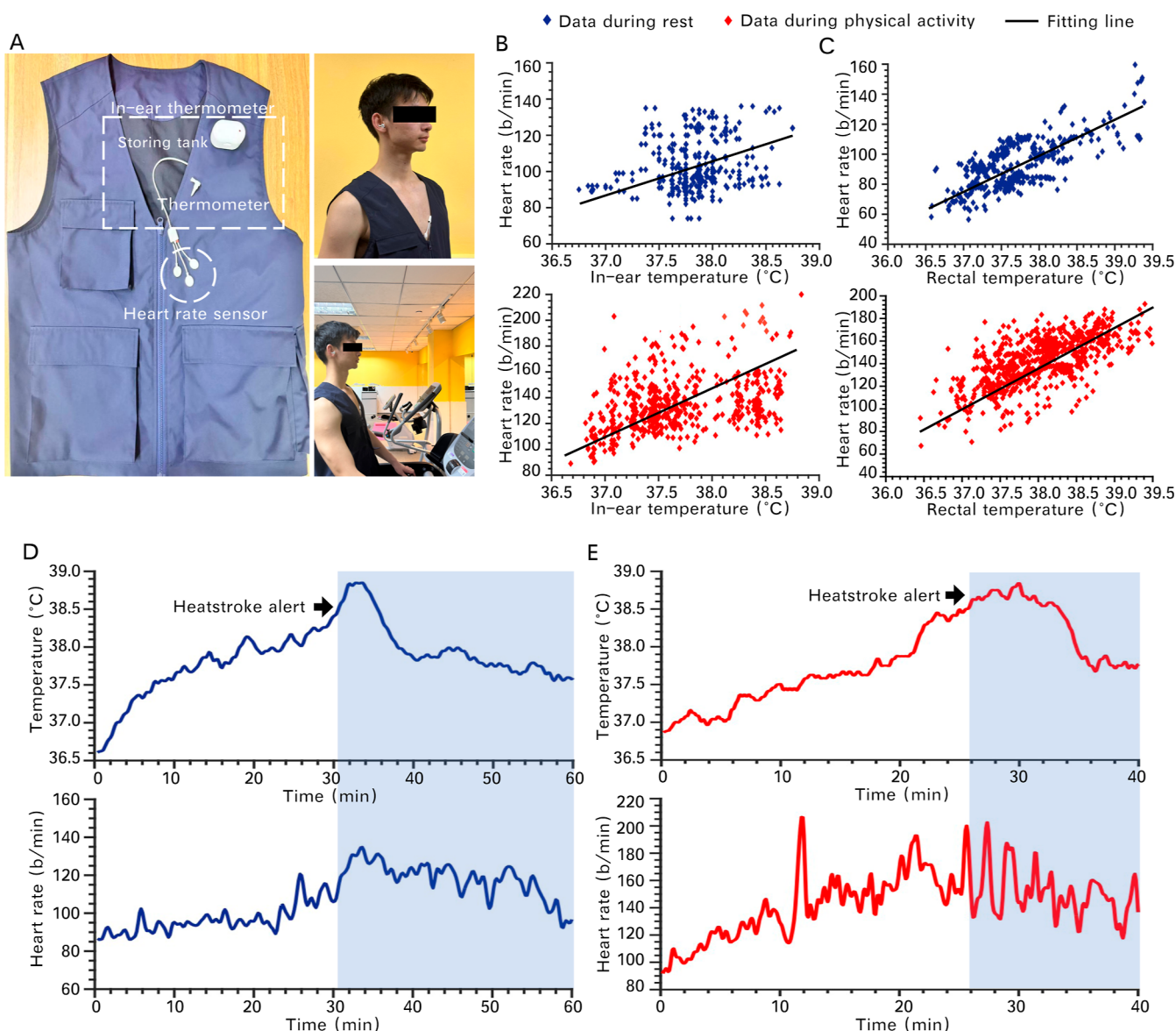


Figure 5. Joint measurements with an in-ear thermometer and a wearable heart rate sensor. (A) The in-ear thermometer, the waistcoat-packaged heart rate sensor, and the experimental conditions; (B,C) shows the heart rate as a function of in-ear temperature in the current experiment, and that as a function of rectal temperature reported by a previous study,³⁰ (C) is reprinted with permission from ref 30, copyright 2015 Oxford University Press; the variations of both temperature and heart rate with heatstroke alert strategy interfered during rest (D) and exercise (E), respectively.

stage, exhibiting a distribution pattern similar to RT but with a narrower range, suggesting that the proposed device achieves a lower measurement deviation for CBT. In contrast, SBT measurements proved to be more sensitive and susceptible to both location-specific and temporal variations. At the experiment's outset, SBT methods, excluding those from the head and thigh, showed substantial discrepancies from rectal CBT. Additionally, these skin-based temperature measurements exhibited wide variability, indicating inherent instability and deviations from CBT. By the end of the experiment, all the tested SBT measurements are statistically distinct from the CBT, indicating that skin temperature can rapidly decrease due to efficient heat dissipation, thereby rendering SBT an inadequate proxy for CBT during physical activity.

Joint Measurements with an In-Ear Thermometer and a Wearable Heart Rate Sensor. A practical experiment was conducted to assess the applicability of the proposed in-ear

thermometer in real-world scenarios. Additionally, a wearable waistcoat-integrated heart rate sensor (detailed in [Supporting Information](#)) was designed to provide supplementary physiological data that could indicate the onset of heatstroke, as illustrated in [Figure 5A](#). The experiment took place in a closed gym environment with controlled conditions of 39 °C temperature and 50% relative humidity, capturing data during both rest periods (exposure to a hot environment without strenuous physical activity) and periods of physical exertion (walking and running on a treadmill). [Figure 5B](#) depicts the potential correlation between heart rate and IET-based CBT during rest and physical activity, with heart rate data in conjunction with rectal-based CBT variation from a previous study (ref 30) as a reference. As indicated in [Figure 5B](#), heart rate demonstrates a rising trend in response to increasing IET when participants are at rest, reaching physiological upper limits of 120–140 beats per minute (bpm) as IET escalates

from 36.5 to 38.5 °C. A similar, albeit more pronounced, trend is observed during physical activities, where heart rate ranges from 80 to 220 bpm as IET increases from 36.5 to 39.0 °C. These findings align with the previous results presented in ref 30, as shown in Figure 5C, suggesting that the elevation in heart rate may be a concurrent physiological response to rising CBT. It is important to acknowledge that establishing a definitive relationship between heart rate and CBT is intricate due to the direct stimulation of the cardiopulmonary system, which independently elevates heart rate. Nonetheless, these experimental observations offer a potential framework for early intervention and alert mechanisms against heat-related injuries such as heatstroke.

Reliable Heatstroke Alert from CBT. Results from this study suggests that CBT can serve as a reliable and direct indicator of heatstroke. Clinically, a rapid escalation of CBT above a threshold of 40.5 °C is recognized as one of the diagnostic criteria for heatstroke. Nevertheless, the practical challenges associated with CBT measurement, such as those conducted rectally, often result in the detection of elevated CBT postheatstroke, thereby delaying diagnosis and increasing the risk of severe outcomes. However, convenient approaches mainly relies on SBTs, which are often inaccurate and unreliable. As illustrated in Figure 4E,F, SBT measurements exhibit significant biases when compared to both rectal CBT and in-ear CBT readings. Leveraging its portability and high accuracy, the proposed in-ear thermometer holds significant potential for the most reliable and convenient early heatstroke detection. In this study, an alerting protocol was established based on the findings from our human subject experiments. As depicted in Figure 4B, the CBT of the participants rose from a normothermic state of approximately 36.5 °C to about 38.5 °C, with subsequent increase becoming more pronounced with extended exercise duration. This observation suggests a potential CBT threshold of 38.5 °C for initiating heatstroke alerts, taking into account individual variations in heat tolerance.

Furthermore, the observed correlation between heart rate and CBT, as illustrated in Figure 5B,C, can also be integrated into the alert strategy. Specifically, a CBT exceeding 38.5 °C triggers a preliminary heatstroke alert, which escalates to an urgent alert if a concurrent increase in heart rate is observed. Figure 5D,E demonstrate the variations in both CBT and heart rate under the implemented alert strategy during rest and physical activity, respectively. In both situations, immediate cooling interventions, such as relocating the volunteers to a cooler environment, applying cold compresses, and encouraging the consumption of cool water, were initiated upon the issuance of a heatstroke alert. Subsequently, CBT rapidly returned to normothermic levels, accompanied by a decline in heart rate. These findings underscore the efficacy of the proposed device in CBT monitoring and its capacity to facilitate early heatstroke alerts.

CONCLUSIONS

This paper introduces a novel, highly compact wearable in-ear thermometer designed for long-term CBT monitoring at the cochlea, with the objective of providing early alerts for heatstroke. The device's miniaturized hardware comprises a negative temperature coefficient sensor embedded in a flexible printed circuit, a miniature microcontroller unit chip with an energy-efficient Bluetooth module, a micro battery, and silicone gel as the encapsulating material. The device measures

27 mm in length by 18 mm in height and weighs a mere 1.3 g, ensuring portability and comfort during wear. Furthermore, an adaptive software system has been developed to support CBT monitoring, data analysis, storage, and device management, offering comprehensive and user-friendly functionalities on the terminal side. To validate the performance of the proposed in-ear thermometers for CBT monitoring accuracy and heatstroke alerting, human subject experiments were conducted, with the following key conclusions:

1. CBT obtained using the proposed in-ear thermometer closely aligns with the gold standard rectal measurement for CBTs below 38.5 °C, demonstrating its accuracy and feasibility in tracking CBT variations during physical activities.
2. For CBTs exceeding 38.5 °C, the in-ear CBTs measured by the device exhibit deviations of up to 0.4 °C compared to rectal measurements, potentially indicating individual variations in heat response.
3. Statistical analysis reveals no significant differences between CBTs from the in-ear thermometer and those from rectal measurements, whereas skin temperature measurements show substantial deviations from rectal measurements, suggesting that surface body temperature is inadequate for reflecting CBT during physical activity.
4. An observed concurrent relationship between heart rate and CBT was noted in both rest and physical activity experiments, potentially offering an additional indicator for early heatstroke detection.
5. The establishment of CBT exceeding 38.5 °C with a continuously increasing heart rate as an urgent heatstroke alert was validated, with results indicating that CBT returned to normothermia within tens of minutes following intervention procedures.

These findings underscore the efficacy of the proposed wearable in-ear thermometer in accurately and continuously monitoring CBT and issuing early alerts for heatstroke, without impeding daily activities, thereby enhancing the prevention and management of heatstroke among populations exposed to heatwaves.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acssensors.4c03470>.

The power consumption and working duration of the proposed device; the operational interface of the software; and the principle of heart rate sensing system (PDF)

AUTHOR INFORMATION

Corresponding Authors

Xue Dong – China-UK Low Carbon College, Shanghai Jiaotong University, Shanghai 20040, China; Email: xue.dong@sjtu.edu.cn

Shuogui Xu – Department of Emergency, Shanghai Changhai Hospital, Shanghai 200433, China; Email: bonexu@139.com

Authors

Wei Zhang – China-UK Low Carbon College, Shanghai Jiaotong University, Shanghai 20040, China; orcid.org/0000-0002-3103-6909

Lei Li – Department of Emergency, Shanghai Changhai Hospital, Shanghai 200433, China
Yimeng Wang – Department of Emergency, Shanghai Changhai Hospital, Shanghai 200433, China
Cenyu Liu – China-UK Low Carbon College, Shanghai Jiaotong University, Shanghai 20040, China
Liyang Sun – China-UK Low Carbon College, Shanghai Jiaotong University, Shanghai 20040, China
Qinglin Guan – China-UK Low Carbon College, Shanghai Jiaotong University, Shanghai 20040, China
Feng Zhang – Huafei Tech Co., Ltd., Suzhou, Jiangsu Province 215512, China

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acssensors.4c03470>

Author Contributions

[†]W.Z., L.L. and Y. W. contributed equally to this work.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This research was funded by the National Key Research and Development Program of China under grant no. 2023YFC3107200, Natural Science Foundation of China under grant no. 52376160, Natural Science Foundation of China under grant no. 52006137, and Postdoctoral Fellowship Program of China Postdoctoral Science Foundation under grant no. GZC20231559.

ABBREVIATIONS

CBT, core body temperature; NTC, negative temperature coefficient; FPC, flexible precision circuit; CHS, classical heatstroke; EHS, exertional heatstroke; RC, resistor capacitance; SoC, system on a chip; MCU, microprogrammed control unit; PWM, power management; ADC, analog to digital converter; USART, universal synchronous/asynchronous receiver/transmitter; GPIO, general-purpose input/output; IET, in-ear temperature; RT, rectal temperature; SBT, surface body temperature

REFERENCES

- (1) Hall, E. J.; Carter, A. J.; O'Neill, D. G. Incidence and risk factors for heat-related illness (heatstroke) in UK dogs under primary veterinary care in 2016. *Sci. Rep.* **2020**, *10* (1), 9128.
- (2) Ogata, S.; Takegami, M.; Ozaki, T.; Nakashima, T.; Onozuka, D.; Murata, S.; Nakaoku, Y.; Suzuki, K.; Hagihara, A.; Noguchi, T.; Iihara, K.; Kitazume, K.; Morioka, T.; Yamazaki, S.; Yoshida, T.; Yamagata, Y.; Nishimura, K. Heatstroke predictions by machine learning, weather information, and an all-population registry for 12-hour heatstroke alerts. *Nat. Commun.* **2021**, *12* (1), 4575.
- (3) Wang, Y.; Li, D.; Wu, Z.; Zhong, C.; Tang, S.; Hu, H.; Lin, P.; Yang, X.; Liu, J.; He, X.; Zhou, H.; Liu, F. Development and validation of a prognostic model of survival for classic heatstroke patients: a multicenter study. *Sci. Rep.* **2023**, *13* (1), 19265.
- (4) Yokoyama, K.; Kaneko, T.; Ito, A.; Ieki, Y.; Kawamoto, E.; Suzuki, K.; Ishikura, K.; Imai, H.; Kanda, J.; Yokobori, S. Sequential organ failure assessment score as a predictor of the outcomes of patients hospitalized for classical or exertional heatstroke. *Sci. Rep.* **2022**, *12* (1), 16373.
- (5) Zhong, L.; Wu, M.; Ji, J.; Liu, Z. Usefulness of sequential organ failure assessment score on admission to predict the 90-day mortality in patients with exertional heatstroke: An over 10-year intensive care survey. *Am. J. Emerg. Med.* **2022**, *61*, 56–60.

- (6) Oka, K.; Honda, Y.; Hui Phung, V. L.; Hijioka, Y. Potential effect of heat adaptation on association between number of heatstroke patients transported by ambulance and wet bulb globe temperature in Japan. *Environ. Res.* **2023**, *216*, 114666.
- (7) Wedler, M.; Pinto, J. G.; Hochman, A. More frequent, persistent, and deadly heat waves in the 21st century over the Eastern Mediterranean. *Sci. Total Environ.* **2023**, *870*, 161883.
- (8) Thompson, V.; Mitchell, D.; Hegerl, G. C.; Collins, M.; Leach, N. J.; Slingo, J. M. The most at-risk regions in the world for high-impact heatwaves. *Nat. Commun.* **2023**, *14* (1), 2152.
- (9) Watts, N.; Amann, M.; Arnell, N.; Ayeb-Karlsson, S.; Beagley, J.; Belesova, K.; Boykoff, M.; Byass, P.; Cai, W.; Campbell-Lendrum, D.; Capstick, S.; Chambers, J.; Coleman, S.; Dalin, C.; Daly, M.; Dasandi, N.; Dasgupta, S.; Davies, M.; Di Napoli, C.; Dominguez-Salas, P.; Drummond, P.; Dubrow, R.; Ebi, K. L.; Eckelman, M.; Ekins, P.; Escobar, L. E.; Georgeson, L.; Golder, S.; Grace, D.; Graham, H.; Haggard, P.; Hamilton, I.; Hartinger, S.; Hess, J.; Hsu, S.-C.; Hughes, N.; Jankin Mikhaylov, S.; Jimenez, M. P.; Kelman, I.; Kennard, H.; Kiesewetter, G.; Kinney, P. L.; Kjellstrom, T.; Kniveton, D.; Lampard, P.; Lemke, B.; Liu, Y.; Liu, Z.; Lott, M.; Lowe, R.; Martinez-Urtaza, J.; Maslin, M.; McAllister, L.; McGushin, A.; McMichael, C.; Milner, J.; Moradi-Lakeh, M.; Morrissey, K.; Munzert, S.; Murray, K. A.; Neville, T.; Nilsson, M.; Sewe, M. O.; Oreszczyn, T.; Otto, M.; Owfi, F.; Pearman, O.; Pencheon, D.; Quinn, R.; Rabbaniha, M.; Robinson, E.; Rocklöv, J.; Romanello, M.; Semenza, J. C.; Sherman, J.; Shi, L.; Springmann, M.; Tabatabaei, M.; Taylor, J.; Triñanes, J.; Shumake-Guillemot, J.; Vu, B.; Wilkinson, P.; Winning, M.; Gong, P.; Montgomery, H.; Costello, A. The 2020 report of The Lancet Countdown on health and climate change: responding to converging crises. *Lancet* **2021**, *397* (10269), 129–170.
- (10) Zhao, Q.; Li, S.; Ye, T.; Wu, Y.; Gasparrini, A.; Tong, S.; Urban, A.; Vicedo-Cabrera, A. M.; Tobias, A.; Armstrong, B.; Royé, D.; Lavigne, E.; de'Donato, F.; Sera, F.; Kan, H.; Schwartz, J.; Pascal, D.; Ryt, N.; Goodman, P.; Saldiva, P. H. N.; Bell, M. L.; Guo, Y.; on behalf of the MCC Collaborative Research Network. Global, regional, and national burden of heatwave-related mortality from 1990 to 2019: A three-stage modelling study. *PLoS Med.* **2024**, *21* (5), No. e1004364.
- (11) Romanello, M.; Napoli, C. d.; Green, C.; Kennard, H.; Lampard, P.; Scamman, D.; Walawender, M.; Ali, Z.; Ameli, N.; Ayeb-Karlsson, S.; Beggs, P. J.; Belesova, K.; Berrang Ford, L.; Bowen, K.; Cai, W.; Callaghan, M.; Campbell-Lendrum, D.; Chambers, J.; Cross, T. J.; van Daalen, K. R.; Dalin, C.; Dasandi, N.; Dasgupta, S.; Davies, M.; Dominguez-Salas, P.; Dubrow, R.; Ebi, K. L.; Eckelman, M.; Ekins, P.; Freyberg, C.; Gasparyan, O.; Gordon-Strachan, G.; Graham, H.; Gunther, S. H.; Hamilton, I.; Hang, Y.; Hänninen, R.; Hartinger, S.; He, K.; Heidecke, J.; Hess, J. J.; Hsu, S.-C.; Jamart, L.; Jankin, S.; Jay, O.; Kelman, I.; Kiesewetter, G.; Kinney, P.; Kniveton, D.; Kouznetsov, R.; Larosa, F.; Lee, J. K. W.; Lemke, B.; Liu, Y.; Liu, Z.; Lott, M.; Lotto Batista, M.; Lowe, R.; Odhiambo Sewe, M.; Martinez-Urtaza, J.; Maslin, M.; McAllister, L.; McMichael, C.; Mi, Z.; Milner, J.; Minor, K.; Minx, J. C.; Mohajeri, N.; Momen, N. C.; Moradi-Lakeh, M.; Morrissey, K.; Munzert, S.; Murray, K. A.; Neville, T.; Nilsson, M.; Obradovich, N.; O'Hare, M. B.; Oliveira, C.; Oreszczyn, T.; Otto, M.; Owfi, F.; Pearman, O.; Pega, F.; Pershing, A.; Rabbaniha, M.; Rickman, J.; Robinson, E. J. Z.; Rocklöv, J.; Salas, R. N.; Semenza, J. C.; Sherman, J. D.; Shumake-Guillemot, J.; Silbert, G.; Sofiev, M.; Springmann, M.; Stowell, J. D.; Tabatabaei, M.; Taylor, J.; Thompson, R.; Tonne, C.; Treskova, M.; Trinanes, J. A.; Wagner, F.; Warnecke, L.; Whitcombe, H.; Winning, M.; Wyns, A.; Yglesias-González, M.; Zhang, S.; Zhang, Y.; Zhu, Q.; Gong, P.; Montgomery, H.; Costello, A. The 2023 report of the Lancet Countdown on health and climate change: the imperative for a health-centred response in a world facing irreversible harms. *Lancet* **2023**, *402* (10419), 2346–2394.
- (12) King, M. A.; Ward, M. D.; Mayer, T. A.; Plamper, M. L.; Madsen, C. M.; Chevront, S. N.; Kenefick, R. W.; Leon, L. R. Influence of prior illness on exertional heat stroke presentation and outcome. *PLoS One* **2019**, *14* (8), No. e0221329.

(13) Mani, G. K.; Nimura, Y.; Tsuchiya, K. Advanced Artificial Electronic Skin Based pH Sensing System for Heatstroke Detection. *ACS Sens.* **2020**, *5* (4), 911–916.

(14) Su, L.; Wang, J.; Liu, B.; Liu, H.; Chen, Q.; Liu, J.; Li, S.; Yuan, L.; An, L.; Lin, H.; Feng, L.; Zheng, J.; Ren, J.; Liang, L.; Li, S. Construction of a Near-Infrared Fluorescent Probe for Dynamic Monitoring and Early Diagnosis of Heart Failure. *ACS Sens.* **2024**, *9* (6), 3075–3084.

(15) Garzón-Villalba, X. P.; Wu, Y.; Ashley, C. D.; Bernard, T. E. Ability to Discriminate Between Sustainable and Unsustainable Heat Stress Exposures—Part 2: Physiological Indicators. *Ann. Work Exposures Health* **2017**, *61* (6), 621–632.

(16) Park, G.; Woo, S.; Kim, K.; Kim, J.; Hwang, J.; Kim, S. K.; Lee, H.; Lee, S.; Kwon, B.; Kim, S.; Rhee, H.; Kim, W. Noninvasive and Continuous Monitoring of the Core Body Temperature through the Quantitative Measurement of Blood Perfusion Rate. *ACS Sens.* **2023**, *8* (8), 2975–2985.

(17) Yaldiz, C. O.; Buller, M. J.; Richardson, K. L.; An, S.; Lin, D. J.; Satish, A.; Driver, K.; Atkinson, E.; Mesite, T.; King, C.; Bursey, M.; Galer, M.; Millard-Stafford, M.; Sawka, M. N.; Medda, A.; Inan, O. T. Early Prediction of Impending Exertional Heat Stroke With Wearable Multimodal Sensing and Anomaly Detection. *IEEE J. Biomed. Health Inform.* **2023**, *27* (12), 5803–5814.

(18) Bariya, M.; Davis, N.; Gillan, L.; Jansson, E.; Kokkonen, A.; McCaffrey, C.; Hiltunen, J.; Javey, A. Resettable Microfluidics for Broad-Range and Prolonged Sweat Rate Sensing. *ACS Sens.* **2022**, *7* (4), 1156–1164.

(19) Li, Y.; Chen, W.; Lu, L. Wearable and Biodegradable Sensors for Human Health Monitoring. *ACS Appl. Bio Mater.* **2021**, *4* (1), 122–139.

(20) Chen, S.-T.; Lin, S.-S.; Lan, C.-W.; Hsu, H.-Y. Design and Development of a Wearable Device for Heat Stroke Detection. *Sensors* **2018**, *18* (1), 17.

(21) Li, F.; Xue, H.; Lin, X.; Zhao, H.; Zhang, T. Wearable Temperature Sensor with High Resolution for Skin Temperature Monitoring. *ACS Appl. Mater. Interfaces* **2022**, *14* (38), 43844–43852.

(22) Huan, J.; Bernstein, J. S.; Difuntorum, P.; Masna, N. V. R.; Gravenstein, N.; Bhunia, S.; Mandal, S. A Wearable Skin Temperature Monitoring System for Early Detection of Infections. *IEEE Sens. J.* **2022**, *22* (2), 1670–1679.

(23) Momose, H.; Takasaka, M.; Watanabe-Asaka, T.; Hayashi, M.; Maejima, D.; Kawai, Y.; Ohhashi, T. Heatstroke risk informing system using wearable perspiration ratemeter on users undergoing physical exercise. *Sci. Rep.* **2023**, *13* (1), 416.

(24) Wang, S.; Rovira, M.; Demuru, S.; Lafaye, C.; Kim, J.; Kunnell, B. P.; Besson, C.; Fernandez-Sanchez, C.; Serra-Graells, F.; Margarit-Taulé, J. M.; Aymerich, J.; Cuenca, J.; Kiselev, I.; Gremeaux, V.; Saubade, M.; Jimenez-Jorquera, C.; Briand, D.; Liu, S. C. Multimensing Wearables for Real-Time Monitoring of Sweat Electrolyte Biomarkers During Exercise and Analysis on Their Correlation With Core Body Temperature. *IEEE Trans. Biomed. Circuits Syst.* **2023**, *17* (4), 808–817.

(25) Chaglla E, J. S.; Celik, N.; Balachandran, W. Measurement of Core Body Temperature Using Graphene-Inked Infrared Thermopile Sensor. *Sensors* **2018**, *18* (10), 3315.

(26) Ellebrecht, D. B.; Gola, D.; Kaschwich, M. Evaluation of a Wearable in-Ear Sensor for Temperature and Heart Rate Monitoring: A Pilot Study. *J. Med. Syst.* **2022**, *46* (12), 91.

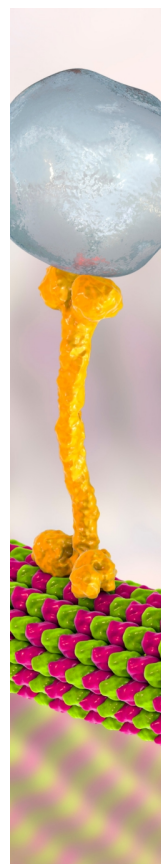
(27) Ota, H.; Chao, M.; Gao, Y.; Wu, E.; Tai, L.-C.; Chen, K.; Matsuoka, Y.; Iwai, K.; Fahad, H. M.; Gao, W.; Nyein, H. Y. Y.; Lin, L.; Javey, A. 3D Printed “Earable” Smart Devices for Real-Time Detection of Core Body Temperature. *ACS Sens.* **2017**, *2* (7), 990–997.

(28) Webster, D. B.; Packer, D. J.; Webster, M. Functional anatomy of the external and middle ear. *Ear Nose Throat J.* **1985**, *64* (6), 275–281.

(29) Li, L.; Chen, J.; Chen, J.; Wang, Y.; Pei, Y.; Wang, M.; Chang, W.; Ma, J.; Song, Q.; Xu, S. Heat stress induces a three-phase

thermoregulatory response in different hot and humid environments in rats. *Sci. Total Environ.* **2024**, *954*, 176476.

(30) Richmond, V. L.; Davey, S.; Griggs, K.; Havenith, G. Prediction of Core Body Temperature from Multiple Variables. *Ann. Occup. Hyg.* **2015**, *59* (9), 1168–1178.



CAS BIOFINDER DISCOVERY PLATFORM™

BRIDGE BIOLOGY AND CHEMISTRY FOR FASTER ANSWERS

Analyze target relationships,
compound effects, and disease
pathways

Explore the platform



A Division of the
American Chemical Society